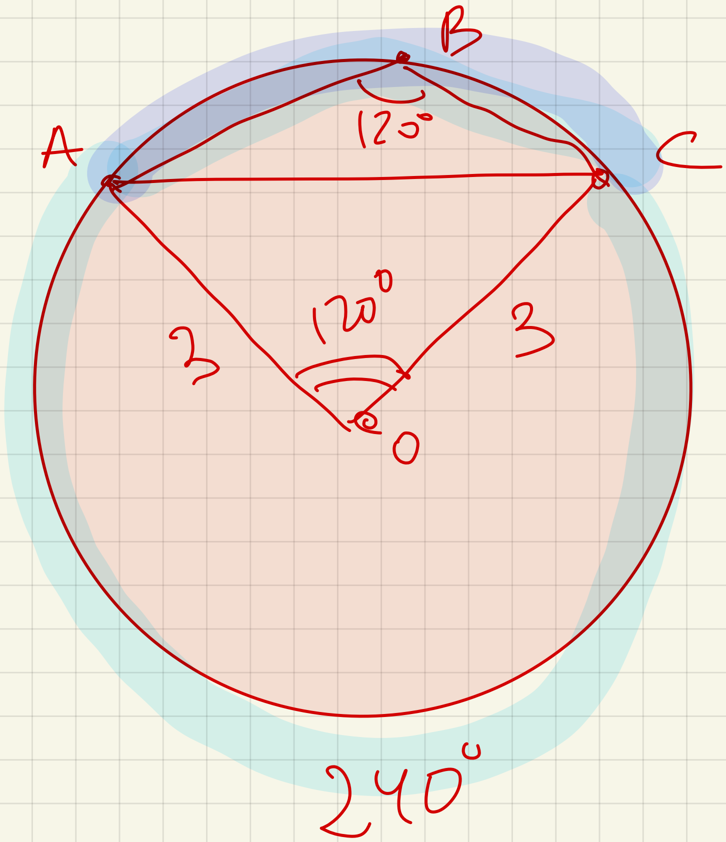
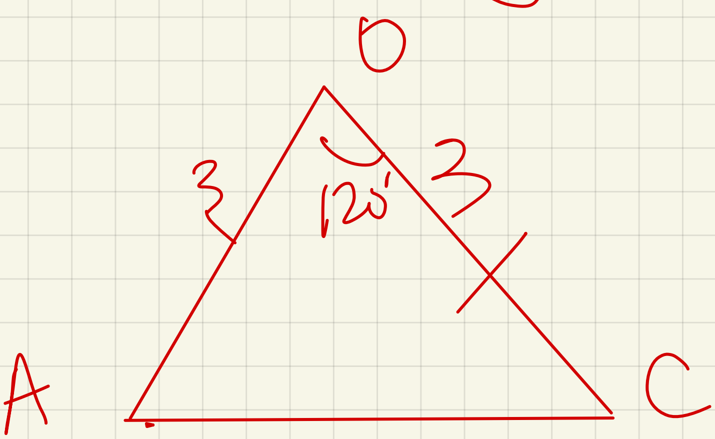
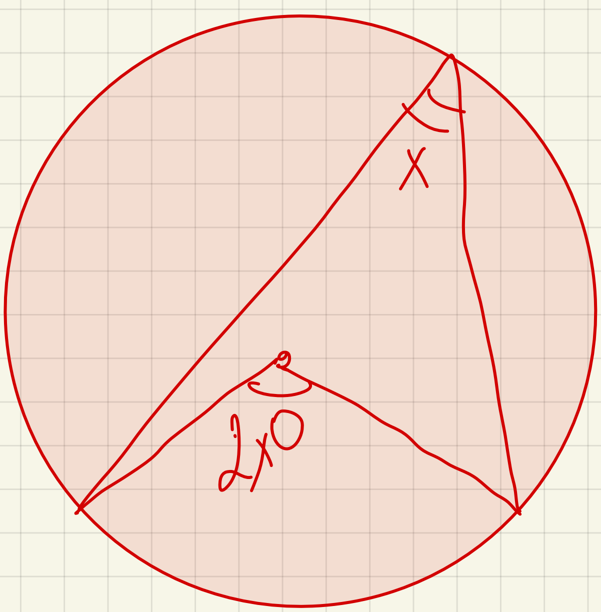


$\cup AC_б = 2 \cdot 120 = 240°$

$\cup AC_м = 120°$





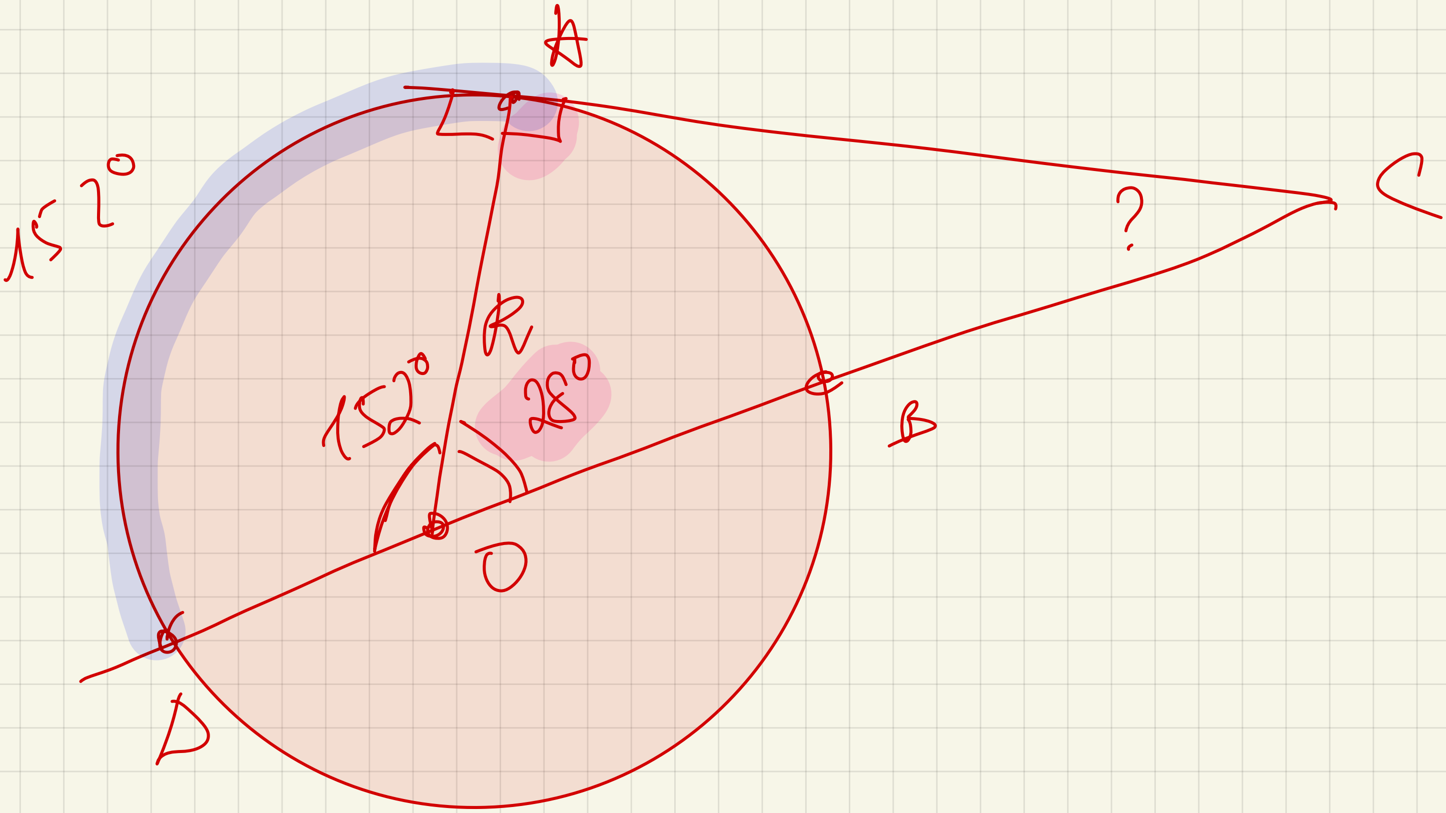
$2x \qquad x$

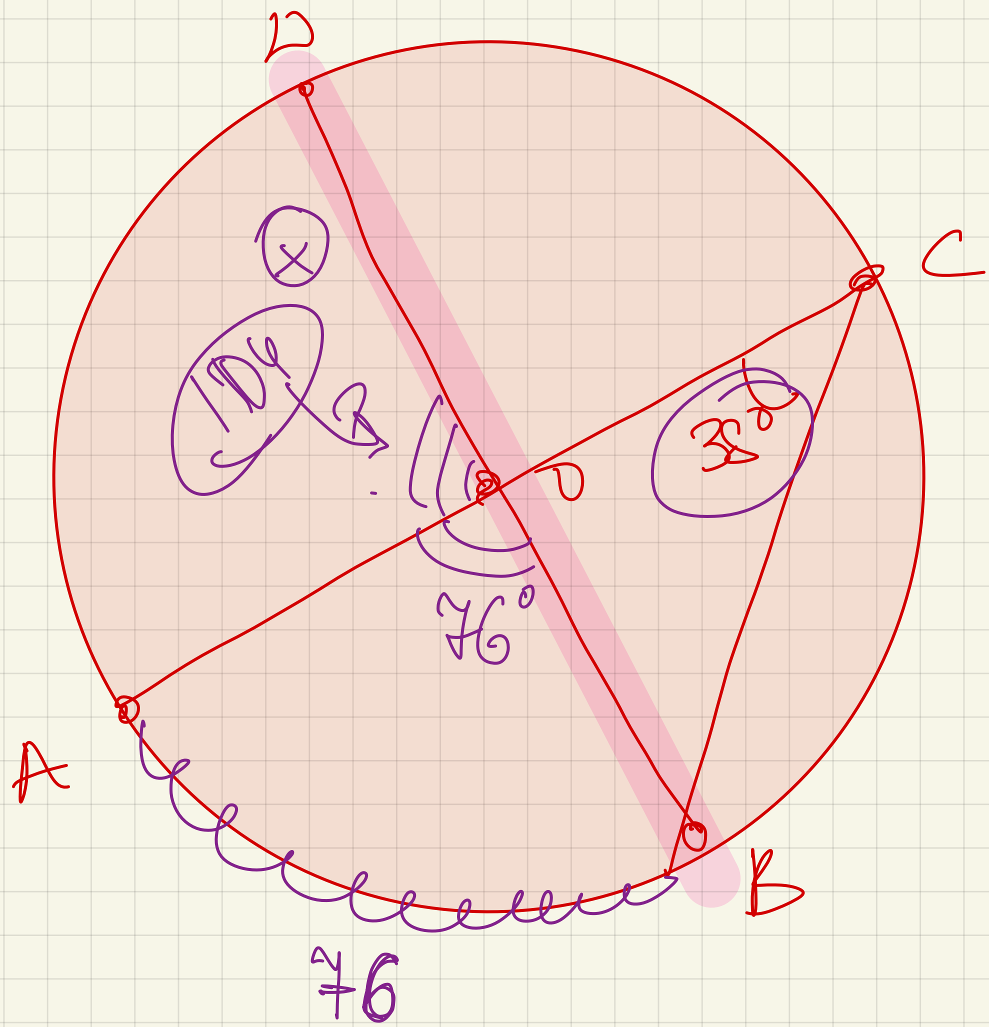
$\cup > B$

на $36°$

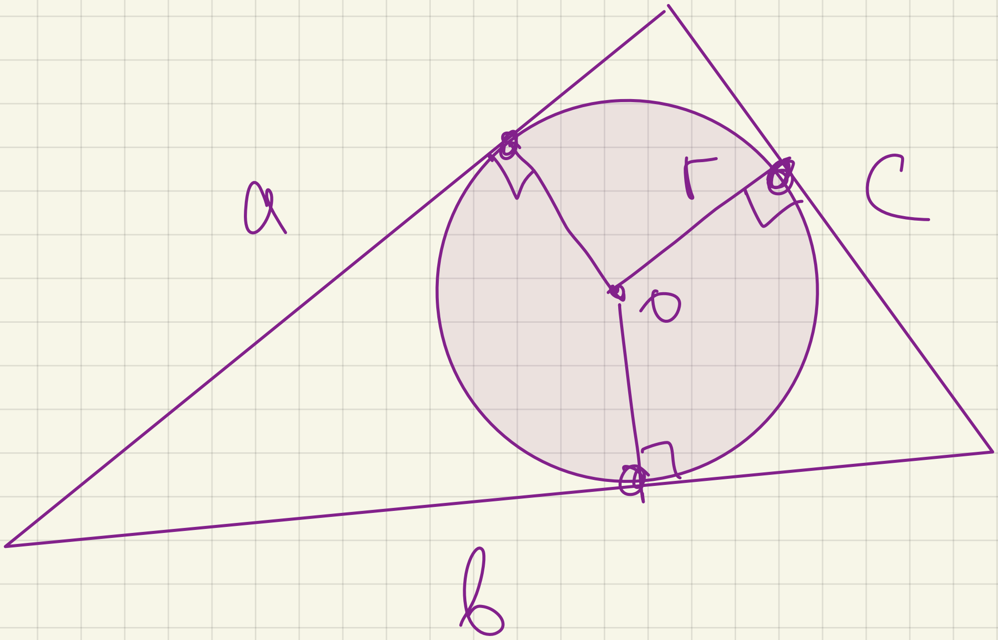
$\underbrace{2x} - \underbrace{x} = 36$

$x = 36$


A
C
152°
?
R
152°
28°
B
O
D


D
x
C
104
38°
O
76°
A
B
76

Вписанная окр.

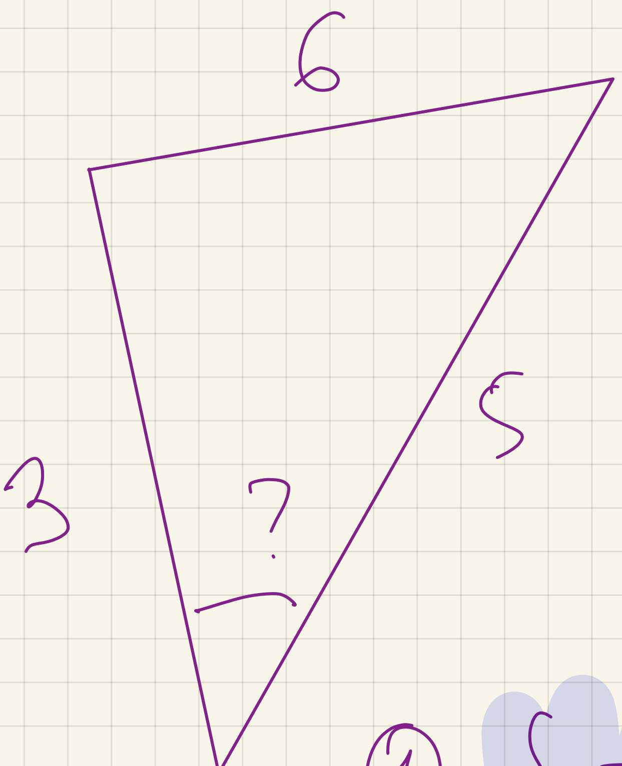


$S = p \cdot r$

$p = \frac{a+b+c}{2}$

$S = \left(\frac{a+b+c}{2}\right) r$

$r = \frac{S}{\frac{a+b+c}{2}}$



r ?

$\frac{12}{8}$?

$S = \frac{2\sqrt{14}}{7}$

① $S = \sqrt{p(p-a)(p-b)(p-c)}$

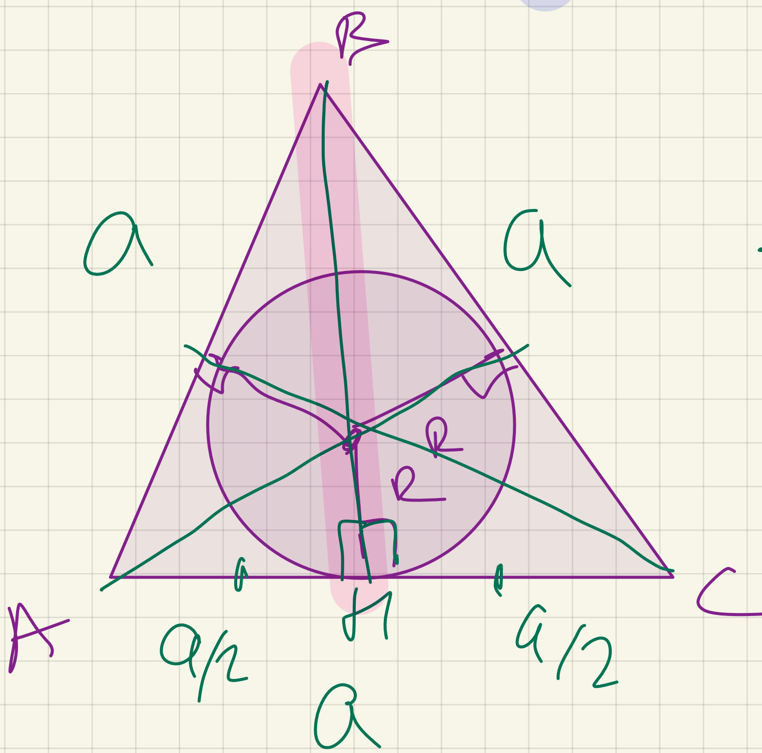
$p = 7$ $\sqrt{7(7-3)(7-5)(7-6)}$

$\sqrt{4 \cdot 4 \cdot 2} = 2\sqrt{14}$?

② теор. cos → $\cos x =$.

$\rightarrow 1-\cos^2 x = \sin^2 x$

$$S = \frac{a\,b\,\sin x}{2}$$



$$S = \frac{BH \cdot AC}{2}$$

$$BH^2 = \sqrt{a^2 - \frac{a^2}{4}}$$

$$\sqrt{\frac{4a^2 - a^2}{4}} =$$

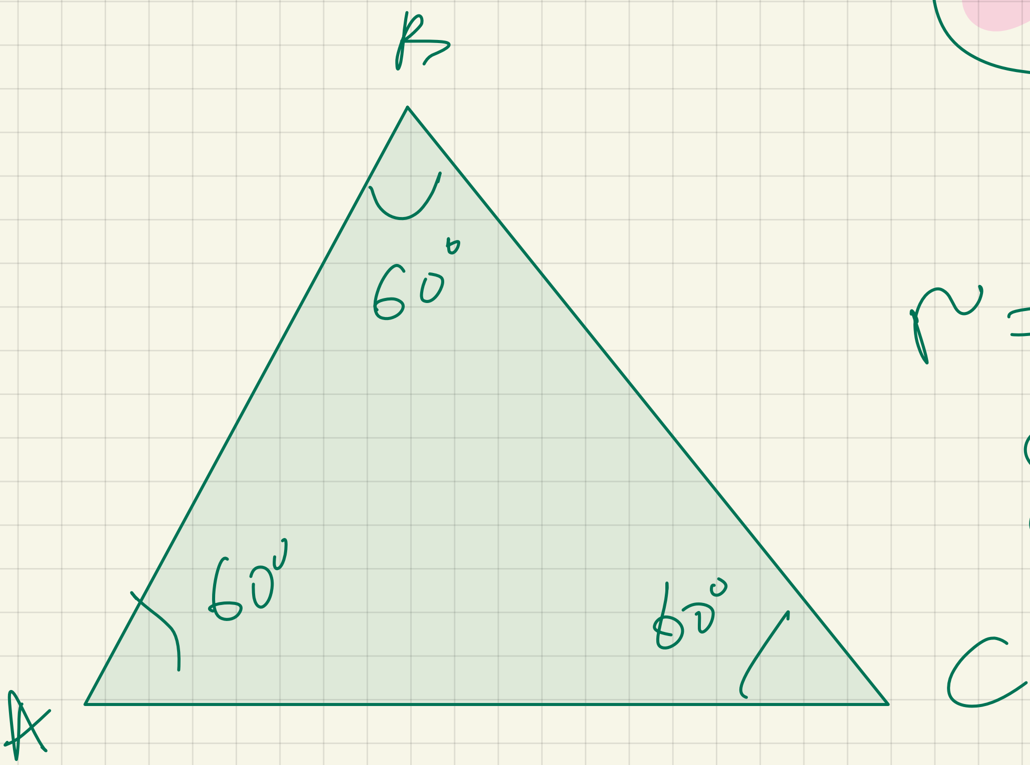
$$\frac{a\sqrt{3}}{2}$$

$$S = \frac{\frac{a\sqrt{3}}{2} \cdot a}{2} = \frac{a^2\sqrt{3}}{4}$$

$$r = \frac{a^2\sqrt{3}}{4} : \frac{3a}{2} =$$

$$= \frac{a^2\sqrt{3}}{\cancel{4}_{2}} \cdot \frac{\cancel{2}}{3a} = \frac{a^{\cancel{2}}\sqrt{3}}{6\cancel{a}} =$$

$$\frac{a\sqrt{3}}{6}$$



$r = 6\sqrt{3}$

$S = ?$

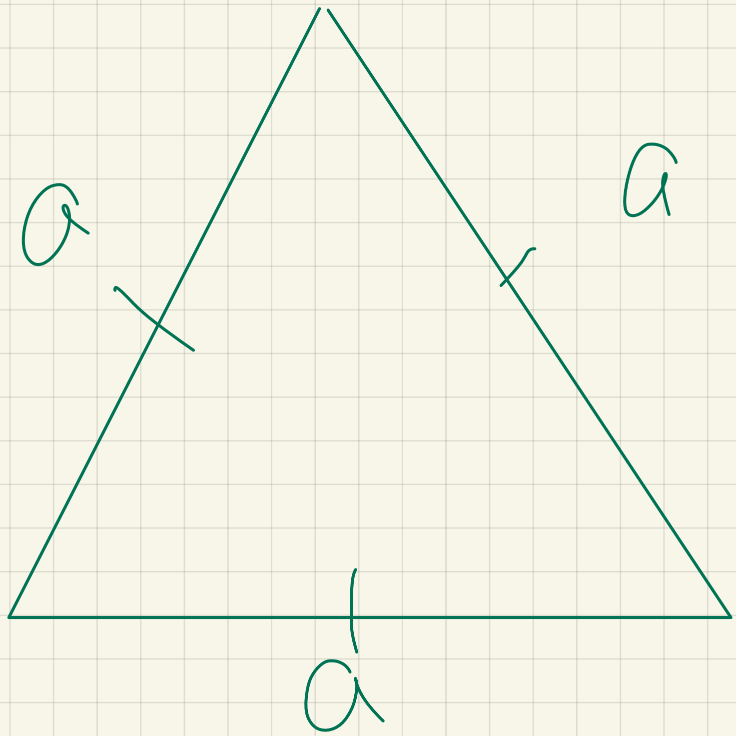
$$r = \frac{a\sqrt{3}}{6}$$

$$S = \frac{1}{2} \cdot a \cdot a \cdot \frac{\sqrt{3}}{2}$$

$$= \frac{36^2\sqrt{3}}{4}$$

$$6\sqrt{3} = \frac{a\sqrt{3}}{6}$$

$$a = \frac{6 \cdot 6\cancel{\sqrt{3}}}{\cancel{\sqrt{3}}} = 36$$



$$p = \frac{3\sqrt{3}}{6}$$

$\rightarrow a\,?$

$r\,?$

$r =$

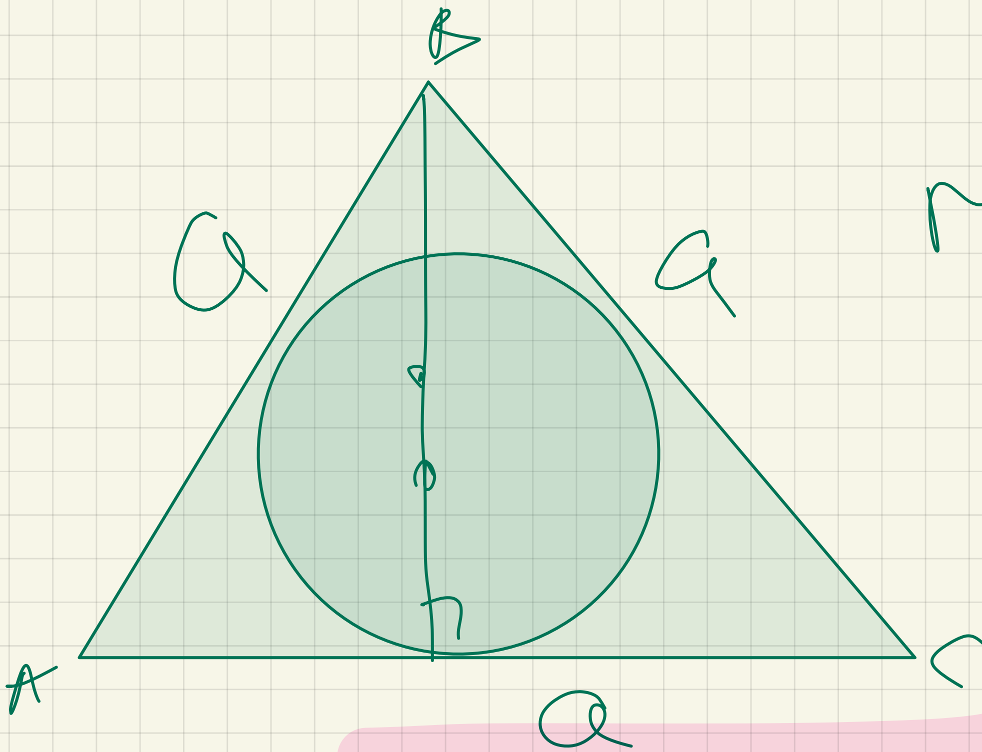
$$p = \frac{\cancel{3}a}{2} = \frac{\cancel{3}\sqrt{3}}{6}$$

$$a = \frac{2\sqrt{3}}{6} = \frac{\sqrt{3}}{3}$$

$$S = \frac{a^2\sqrt{3}}{4} = \frac{\cancel{3}}{\cancel{9}} \cdot \frac{\sqrt{3}}{4} = \frac{\sqrt{3}}{12}$$

$$\frac{S}{p} = r \Rightarrow \quad r = \frac{\sqrt{3}}{12} : \frac{3\sqrt{3}}{6} =$$

$$\frac{\sqrt{3}\cdot \cancel{6}}{\cancel{12}_{2}\cdot 3\cancel{\sqrt{3}}} = \frac{1}{6}$$



$r = 6$

h ?

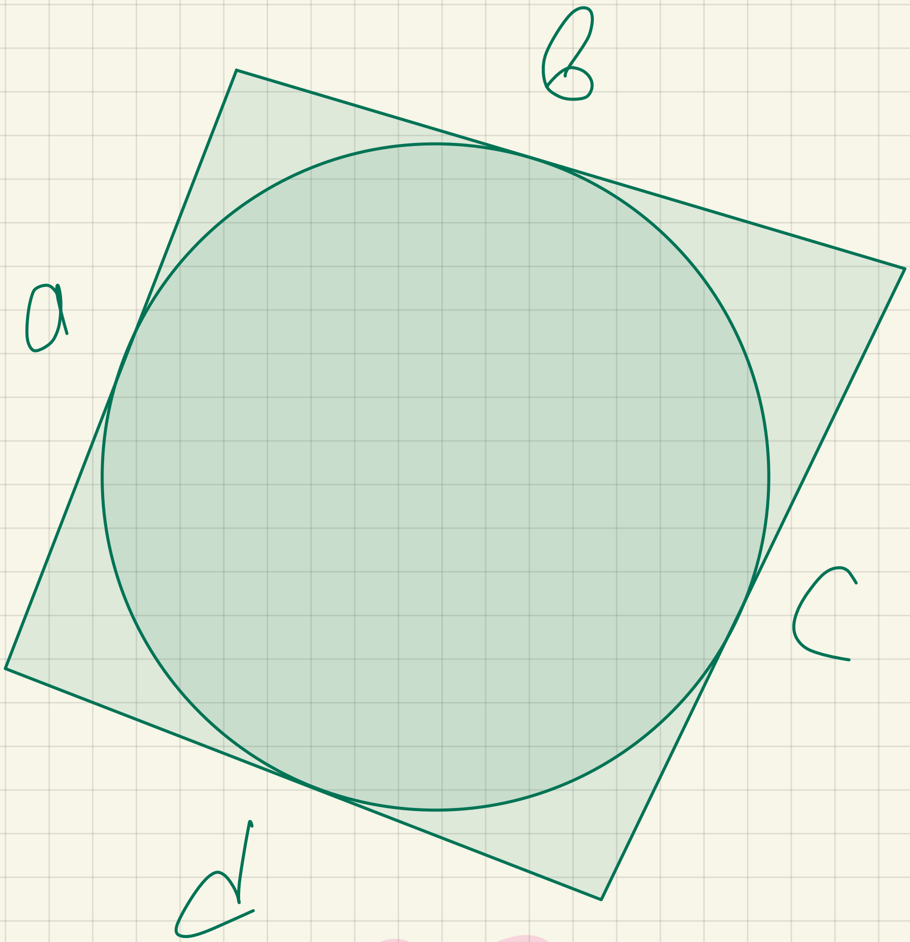
$$S = \boxed{p \cdot r = \frac{h \cdot a}{2}}$$

$$p = \frac{3a}{2}$$

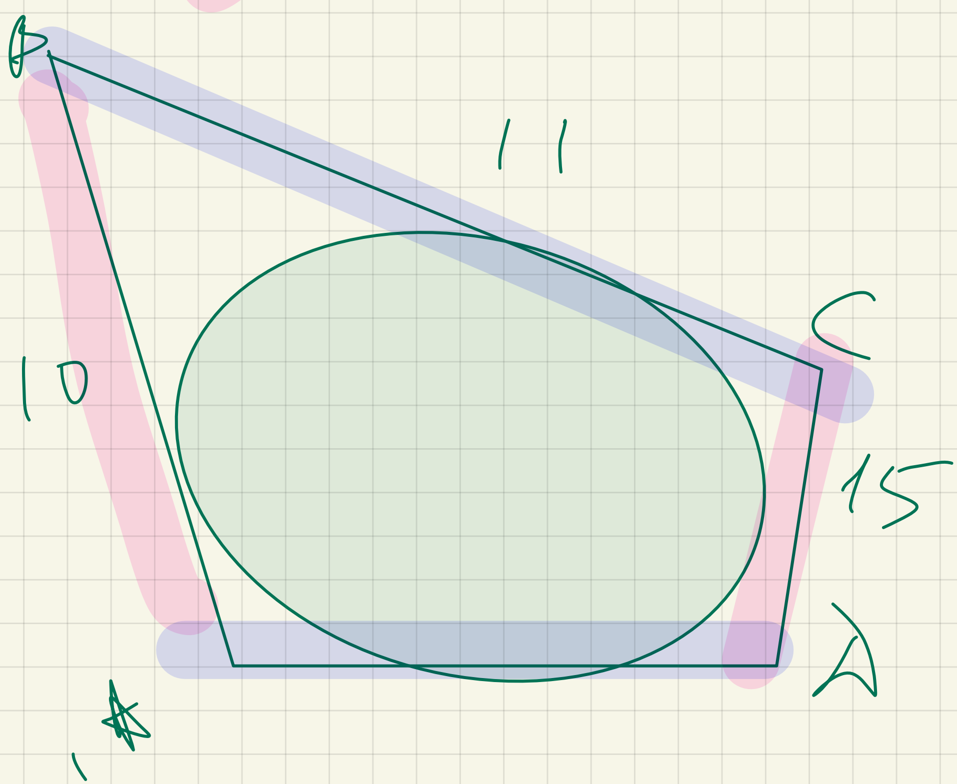
$$\frac{3\cancel{a}}{\cancel{2}} r = \frac{h \cdot \cancel{a}}{\cancel{2}}$$

$$r = \frac{h}{3}$$

$$h = 3r = 18$$



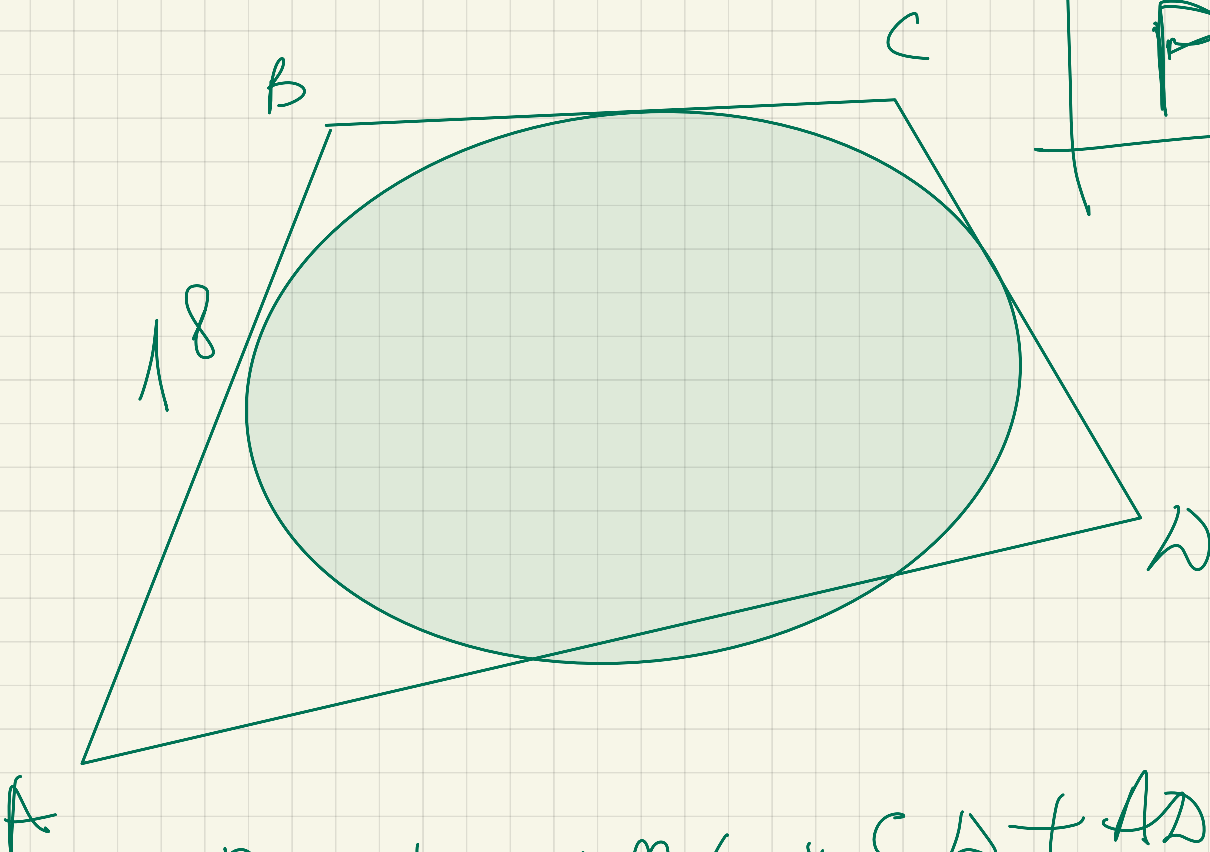
$a + c = b + d$ → можно вписать окруж



AD?

$10 + 15 = 11 + AD$

$AD = 14$



$P = 54$

CD?

$P = AB + BC + CD + AD = 54$

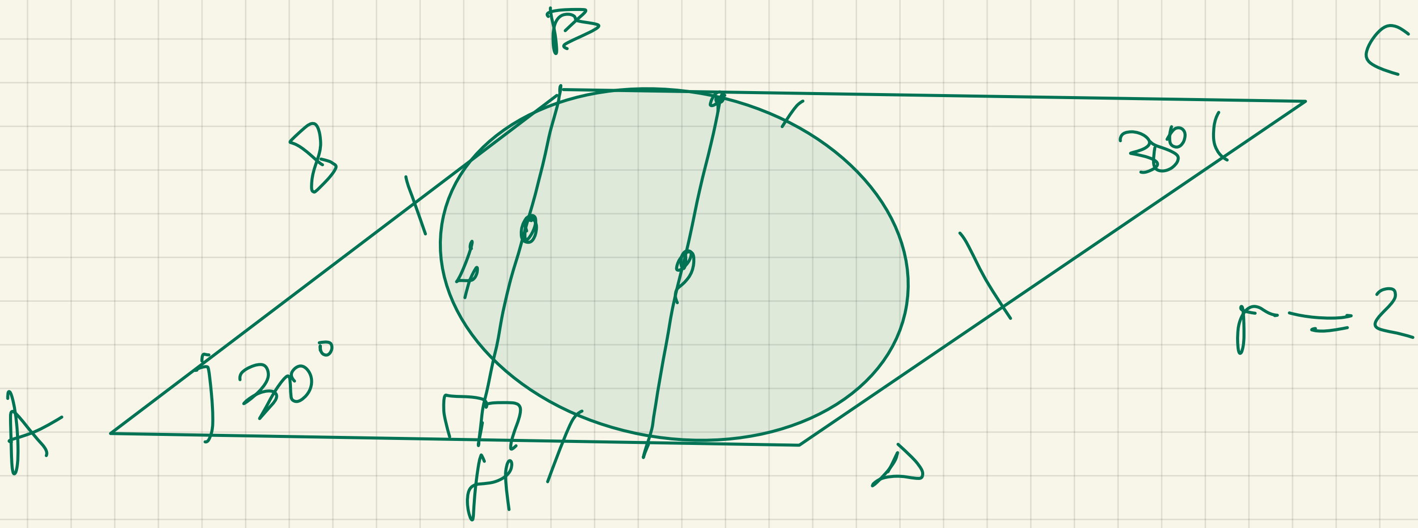
$\underbrace{AB + CD} = BC + AD$

$AB + CD + AB + CD =$

$2(AB + CD) = 54$

$AB + CD = 27$

$CD = 27 - 18 = 9$



$BH = 2r$

$BH = 4$